14E022 Macroeconomics I

Practice Exam 3

BSE — Academic Year 2025–26

*Instructions: the exam is made of 3 problems and a set of short questions totalling **100 points**. You have **2 hours**. Show all derivations clearly. For short questions, concise and precise answers score highest.*

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1 Question 1 — Romer (1990) Expanding Varieties (25 points)

Consider a three-sector economy consisting of a final goods sector, an intermediate goods sector, and an R&D sector. The representative final goods firm produces output using production workers L_Y and a continuum of differentiated intermediate inputs:

$$Y_t = L_Y^{1-\alpha} \int_0^{M_t} x_i^{\alpha} \, \mathrm{d}i, \qquad \alpha \in (0,1),$$

where M_t is the total mass of available varieties and x_i is the quantity of variety i used. Each intermediate variety is produced one-for-one from the final good (marginal cost = 1). The R&D technology expands the variety frontier at rate

$$\dot{M}_t = \lambda M_t L_{A,t},$$

where $L_{A,t}$ is the mass of workers employed in R&D and $\lambda > 0$ is a productivity parameter. Households have CRRA preferences with risk-aversion $\sigma > 0$, discount rate $\rho > 0$, and there is no population growth. Total labour supply is $L = L_Y + L_A$.

1. (5 points) Final goods sector FOC.

(a) Set up the final goods firm's profit-maximisation problem taking output as the numéraire and each p_i as given. Derive the isoelastic inverse demand for variety i:

$$p_i = \alpha L_Y^{1-\alpha} \, x_i^{\alpha-1}.$$

(b) Invert this to obtain the demand function $x_i(p_i)$ and show that the price elasticity of demand is $\varepsilon = 1/(1-\alpha) > 1$.

2. (6 points) Intermediate monopolist's pricing and profits.

Each variety i is supplied by a monopolist who bought the blueprint from the R&D sector.

(a) Using the demand function from part 1 and the CES markup formula $p = MC \cdot \varepsilon/(\varepsilon - 1)$, derive the monopoly price $p_i^* = 1/\alpha$. (You may alternatively derive it directly from the monopolist's FOC.)

(b) Substitute $p_i^* = 1/\alpha$ into the demand function to obtain the equilibrium quantity:

$$x_i^* = \alpha^{2/(1-\alpha)} \, L_Y.$$

(c) Show that equilibrium flow profit is

$$\pi = (p_i^* - 1) \, x_i^* = (1-\alpha) \, \alpha^{\alpha/(1-\alpha)} \, L_Y.$$

Verify that $\pi = \frac{1-\alpha}{\alpha} \, x_i^*$.

3. (7 points) R&D equilibrium and the BGP growth rate.

(a) **No-arbitrage (Bellman equation).** Let V_t be the value of owning one blueprint (one variety). Unlike quality ladders, there is no creative destruction in Romer: once created, a variety exists forever. Write the no-arbitrage condition $rV = \pi + \dot{V}$. On the BGP where V is constant, show this simplifies to $rV = \pi$.

(b) **R&D free entry.** A researcher can create a new variety by hiring labour at wage w. The cost of discovering one new variety is $w/(\lambda M)$ (the reciprocal of the marginal product of research labour). Free entry drives profits from R&D to zero:

$$V = \frac{w}{\lambda M}.$$

Combine with the no-arbitrage condition to show $r = \lambda M \pi / w$.

(c) **Labour market and Euler equation.** The wage equals the marginal product of L_Y in the final sector: $w = (1-\alpha)Y/L_Y$. Use the resource allocations $L_Y + L_A = L$ and the Euler equation $\dot{\tilde{c}}/\tilde{c} = (r-\rho)/\sigma$ to derive the BGP growth rate:

$$g^{\text{mkt}} = \frac{\lambda(1-\alpha)L - \rho}{\sigma + 1}.$$

Identify the **scale effect**: explain why g^{mkt} rises with total labour supply L.

Hint:

On the BGP, L_Y and L_A are constant fractions of L. Use the free-entry condition and wage expression to pin down L_Y^*, substitute into the Euler equation.

4. (7 points) Social planner vs. market.

(a) Explain the **two distortions** in the decentralised equilibrium:

(i) *Markup distortion*: the monopolist sets $p_i = 1/\alpha > 1 = MC$, so each variety is under-used relative to the social optimum. How does the planner set quantity?

(ii) *Knowledge spillover*: when a new variety is created, the inventor captures only the discounted flow of monopoly profits V, but the social value also includes consumer surplus and the indirect effect of variety expansion on the productivity of future R&D ($\partial \dot{M}/\partial M$). State the direction of each distortion (under- or over-provision).

(b) Show that $g^{\text{plnr}} > g^{\text{mkt}}$. You do not need to derive g^{plnr} explicitly; it suffices to argue that the planner uses more of each variety (correcting the markup) and internalises the spillover, both of which raise the incentive to conduct R&D.

(c) Identify the **optimal policy mix** that implements the first-best:

(i) A subsidy $s_x = 1 - \alpha$ to intermediate goods purchasers (final goods firms pay only α per unit, eliminating the markup wedge).

(ii) An R&D subsidy that internalises the knowledge-spillover externality.

Argue that a single instrument *cannot* achieve the first-best because the two distortions are independent.

2 Question 2 — Structural Change: Baumol Effect and Non-Homothetic Preferences (25 points)

Consider a two-sector economy with manufacturing (M) and services (S). Production technologies are linear in labour:

$$Y_M = A_M L_M, \qquad Y_S = A_S L_S,$$

with productivity dynamics $\dot{A}_M/A_M = g > 0$ and $\dot{A}_S/A_S = 0$ (services are the stagnant sector). Labour markets are competitive and the total labour force is fixed: $L_M + L_S = L$.

1. (5 points) Competitive factor markets and relative prices.

(a) With a competitive labour market, wages are equalised across sectors. Show that profit maximisation in each sector implies $p_M A_M = p_S A_S = w$. Hence derive:

$$\frac{p_S}{p_M} = \frac{A_M}{A_S}.$$

(b) Explain why this relative price grows at rate g over time. What is the economic interpretation? This is the **cost disease** (Baumol, 1967): wages in the stagnant sector must keep pace with the productive sector, pushing up services prices.

2. (7 points) Baumol mechanism with CES preferences.

Households have CES preferences:

$$U = \left[\omega\, Y_M^{\rho} + (1-\omega)\, Y_S^{\rho}\right]^{1/\rho}, \qquad \rho \in (-\infty, 1) \setminus \{0\},$$

with elasticity of substitution $\sigma_s = 1/(1-\rho)$.

(a) Derive the expenditure-optimal consumption ratio from the household's static optimum:

$$\frac{Y_S}{Y_M} = \left[\frac{(1-\omega)}{\omega} \cdot \frac{p_M}{p_S}\right]^{\sigma_s}.$$

(b) Using $Y_j = A_j L_j$, convert the output ratio to an employment ratio:

$$\frac{L_S}{L_M} = \left(\frac{1-\omega}{\omega}\right)^{\sigma_s} \left(\frac{A_M}{A_S}\right)^{1-\sigma_s}.$$

(c) Analyse the dynamics of L_S/L_M over time (as A_M grows):

(i) If $\sigma_s = 1$ (Cobb-Douglas): show employment shares are **constant** despite rising p_S.

(ii) If $\sigma_s < 1$ (complements): show L_S/L_M rises over time, so labour flows to the stagnant sector (**Baumol result**).

(iii) If $\sigma_s > 1$ (substitutes): show labour flows to the productive sector.

(d) Show that aggregate productivity growth satisfies $\gamma_Y = g \cdot (L_M/L)$ and explain why structural change implies $\gamma_Y \to 0$ as $L_M/L \to 0$ (the Baumol growth drag).

3. (6 points) Non-homothetic preferences.

Suppose instead that households have Stone-Geary utility:

$$U = \log(Y_M - \bar{c}_M) + \log(Y_S), \qquad \bar{c}_M > 0,$$

where $\bar{c}_M$ is a subsistence level of manufactured consumption. Total nominal income is $I = p_M Y_M + p_S Y_S$.

(a) Derive the demand functions by maximising U subject to the budget constraint. Show:

$$Y_M = \bar{c}_M + \frac{I - p_M \bar{c}_M}{2\, p_M}, \qquad Y_S = \frac{I - p_M \bar{c}_M}{2\, p_S}.$$

(b) Derive the **income elasticity of demand** for manufacturing:

$$\varepsilon_M^I = \frac{\partial Y_M}{\partial I} \cdot \frac{I}{Y_M} = 1 - \frac{\bar{c}_M p_M}{I - \bar{c}_M p_M} \cdot \frac{p_M \bar{c}_M}{p_M Y_M}.$$

Show that $\varepsilon_M^I < 1$, i.e. manufacturing is a **necessity**.

(c) As income I rises (with prices fixed), show that the expenditure share of manufacturing falls. This is **demand-side structural change** that occurs even with equal productivity growth across sectors.

4. (7 points) Synthesis and empirical comparison.

(a) In the Baumol model (part 2), structural change occurs even with **homothetic preferences** provided $\sigma_s < 1$. What is the key supply-side mechanism? Why is the elasticity of substitution the critical parameter?

(b) In the non-homothetic model (part 3), structural change occurs even with **equal productivity growth** across sectors. What is the key demand-side mechanism? What does the income elasticity below unity imply for how households allocate their marginal dollar as they grow richer?

(c) Duarte and Restuccia (2010) document average annual TFP growth rates of approximately 3.8% in agriculture, 2.4% in manufacturing, and 1.3% in services over 1956–2004 across a broad panel of countries. Is this differential consistent with the Baumol cost-disease mechanism? In the data, employment shares in services rise and manufacturing employment falls as countries grow richer. Based on this evidence, discuss the relative importance of supply-side (Baumol) versus demand-side (non-homothetic) channels for understanding structural change.

3 Question 3 — Solow Model: Golden Rule, Convergence Speed, and Dynamic Inefficiency (25 points)

Consider the standard Solow growth model in continuous time. Aggregate production is

$$Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}, \qquad \alpha \in (0,1),$$

with saving rate $s \in (0,1)$, population growth $\dot{L}/L = n > 0$, TFP growth $\dot{A}/A = g > 0$, and capital depreciation $\delta > 0$.

1. (5 points) Per-effective-worker dynamics.

(a) Define per-effective-worker variables $\tilde{k} \equiv K/(AL)$ and $\tilde{y} \equiv Y/(AL) = \tilde{k}^{\alpha}$. Derive the law of motion:

$$\dot{\tilde{k}} = s\,\tilde{k}^{\alpha} - (n+g+\delta)\,\tilde{k}.$$

(b) Show using the Inada conditions ($f'(\tilde{k}) \to \infty$ as $\tilde{k} \to 0$, $f'(\tilde{k}) \to 0$ as $\tilde{k} \to \infty$) that a unique interior steady state $\tilde{k}^* > 0$ exists. Solve for it:

$$\tilde{k}^* = \left[\frac{s}{n+g+\delta}\right]^{1/(1-\alpha)}.$$

2. (7 points) Golden Rule capital accumulation.

(a) At the steady state $\tilde{k}^*$, express per-effective-worker consumption as:

$$\tilde{c}^* = f(\tilde{k}^*) - (n+g+\delta)\,\tilde{k}^* = \tilde{k}^{*\alpha} - (n+g+\delta)\,\tilde{k}^*.$$

(Note: $\tilde{c}^* = (1-s)\tilde{k}^{*\alpha}$; both expressions are equivalent at the steady state.)

(b) To find the **Golden Rule saving rate** that maximises $\tilde{c}^*$, treat $\tilde{k}^*$ as a free variable (each s pins down a different $\tilde{k}^*$). Differentiate $\tilde{c}^*$ with respect to $\tilde{k}^*$ and set to zero:

$$\frac{\partial \tilde{c}^*}{\partial \tilde{k}^*} = 0 \implies f'(\tilde{k}^*_{\mathrm{GR}}) = n+g+\delta.$$

For Cobb-Douglas, solve explicitly: $\alpha\,(\tilde{k}^*_{\mathrm{GR}})^{\alpha-1} = n+g+\delta$. Show that the corresponding saving rate is $s_{\mathrm{GR}} = \alpha$.

(c) Interpret the Golden Rule. What happens to steady-state consumption if $s > s_{\mathrm{GR}}$? If $s < s_{\mathrm{GR}}$? Define the concept of **dynamic inefficiency** (over-saving) and explain why a Pareto improvement is possible when $s > s_{\mathrm{GR}}$.

3. (6 points) Speed of convergence.

(a) Linearise the law of motion $\dot{\tilde{k}} = s\tilde{k}^{\alpha} - (n+g+\delta)\tilde{k}$ around $\tilde{k}^*$. Define the log-deviation $\hat{k}_t \equiv \ln(\tilde{k}_t/\tilde{k}^*)$ and show that:

$$\dot{\hat{k}}_t \approx -\beta\,\hat{k}_t, \qquad \text{where} \quad \beta = (1-\alpha)(n+g+\delta).$$

This implies $\hat{k}_t = \hat{k}_0\,e^{-\beta t}$: the gap to steady state shrinks exponentially at rate β.

(b) Calibrate the formula. With $\alpha = 1/3$, $n = 0.01$, $g = 0.02$, $\delta = 0.05$, compute β. The empirical estimate from cross-country regressions (Mankiw, Romer, Weil 1992) is approximately 2% per year. Why is the empirical estimate slower than the basic calibration? (*Hint*: human capital augmentation effectively raises the capital share toward 2/3.)

(c) How does β depend on α? Provide economic intuition: why does a higher capital share imply *slower* convergence to the steady state?

4. (7 points) Dynamic inefficiency.

(a) Define **dynamic inefficiency** formally: the steady state is dynamically inefficient if $r^* < n+g$, where $r^* = \alpha \tilde{k}^{*\alpha-1} - \delta$ is the net return to capital. Show that this condition is equivalent to $s > s_{\text{GR}} = \alpha$ using the expression for $\tilde{k}^*$ and the fact that $r^* K = \alpha Y$ at the steady state.

(b) **Abel et al. (1989) test.** A sufficient condition for dynamic *efficiency* in a stochastic economy is that gross capital income exceeds gross investment:

$$r^* K \;\geq\; \delta K + \dot{K}, \quad \text{i.e.,} \quad \alpha Y \;\geq\; (n+g+\delta)K.$$

Abel et al. document that this inequality holds for all G7 economies throughout the post-war period. What does this imply about the empirical relevance of dynamic inefficiency?

(c) Can the **Ramsey model** be dynamically inefficient? Explain rigorously. Specifically, show that the Transversality Condition (TVC):

$$\lim_{t\to\infty} e^{-\rho t}\, \lambda_t\, k_t = 0$$

implies $r^* > n+g$ on the BGP (where λ_t is the costate on capital in per-capita terms and $\rho > n$ by assumption for bounded utility). Contrast this with the Solow model, where s is exogenous and may inadvertently push the economy into the over-saving region.

4 Short Questions (25 points)

For these questions, short and sharp answers will receive the highest marks. Aim for 3–6 sentences per sub-question unless otherwise indicated.

1. (9 points) Kaldor Facts and Balanced Growth.

(a) State **five** Kaldor (1961) stylized facts about long-run economic growth in developed economies. Which family of production functions is consistent with all five facts on a Balanced Growth Path (BGP)? Explain *why* this family is special.

(b) Show formally that a Cobb-Douglas production function $Y = K^{\alpha}(AL)^{1-\alpha}$ implies a **constant capital income share** $\alpha = rK/Y$ regardless of the levels of K, L, or A. Relate this property to why Solow + Cobb-Douglas generates a BGP with constant growth rates of all per-worker variables.

(c) Karabarbounis and Neiman (2014) document a declining global labour share since the 1980s. Does this violation of the constant-shares Kaldor fact invalidate the Solow framework? Briefly describe one model modification (e.g. a non-unit elasticity of substitution between capital and labour) that could accommodate a trending labour share while preserving balanced growth.

2. (8 points) Knowledge Spillovers and the ϕ Parameter.

Consider the general R&D technology

$$\dot{M}_t = \lambda\, M_t^{\phi}\, L_{A,t}, \qquad \lambda > 0.$$

(a) Derive the BGP growth rate $g_M \equiv \dot{M}/M$ for three cases: (i) $\phi = 1$ (Romer 1990); (ii) $\phi < 1$ (Jones 1995 semi-endogenous growth); (iii) $\phi = 0$ (no knowledge spillovers). In each case, what is the role of **population growth** n in sustaining positive long-run growth in M?

(b) "*Standing on shoulders*" (past knowledge raises the productivity of current researchers) vs. "*fishing out*" (the stock of easy discoveries is depleted by past R&D): which value of ϕ corresponds to each metaphor? What empirical evidence from patent citation studies informs the plausible range of ϕ?

(c) An R&D subsidy raises L_A. In which case(s) does this permanently raise the **long-run growth rate** g_M? In which case(s) does it affect only the **level** of the BGP path (transitional effect)? State the key equation that distinguishes the two outcomes.

3. (8 points) Consumption Tax vs. Capital Income Tax in the Ramsey Model.

Consider a Ramsey economy where the government levies a **proportional consumption tax** $\tau_c \in (0, 1)$, returning all revenue as a lump-sum transfer T_t to households. The household's flow budget constraint is:

$$\dot{a}_t = r_t a_t + w_t + T_t - (1 + \tau_c)\, c_t,$$

where a_t denotes assets, and households maximise $\int_0^{\infty} e^{-\rho t} u(c_t)\, \mathrm{d}t$ with $u(c) = c^{1-\sigma}/(1-\sigma)$.

(a) Set up the *current-value* Hamiltonian with costate variable μ_t. Derive the optimality conditions and eliminate μ_t to obtain the **Euler equation** for $\dot{c}/c$.

(b) Show that the consumption tax **does not distort** the intertemporal allocation of consumption: the Euler equation is *identical* to the no-tax case. Provide intuition: why does taxing consumption at a constant proportional rate leave the intertemporal price ratio $c_{t+\mathrm{d}t}/c_t$ unchanged?

(c) Contrast with the **capital income tax** τ_K studied in Practice Exam 1, which replaces r with $r(1-\tau_K)$ in the Euler equation. Explain the asymmetry: why does τ_K distort saving while τ_c does not, even though both taxes reduce household resources? What is the implication of this asymmetry for **optimal tax theory** (e.g., the Chamley-Judd result)?

End of Practice Exam 3. Good luck!

Appendix: Key Formulas Permitted in Exam

The following results may be used without re-derivation unless the question explicitly asks you to derive them.

A.1 Log-differentiation

If $Z_t = X_t^a Y_t^b$, then $\gamma_Z = a\gamma_X + b\gamma_Y$.

A.2 Ramsey / Euler equation (standard, no tax)

$$\frac{\dot{c}}{c} = \frac{1}{\sigma}(r - \rho), \qquad \frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{1}{\sigma}(r - \delta - \rho - \sigma x),$$

where $r = f'(\tilde{k}) - \delta$, $x = \dot{A}/A$, and n is population growth.

A.3 Hamiltonian first-order conditions (current-value)

Control u, state x, costate λ, discount ρ:

$$H_u = 0, \qquad \dot{\lambda} = \rho\lambda - H_x.$$

A.4 Romer (1990) demand function and monopoly markup

Final sector: $Y = L_Y^{1-\alpha} \int_0^M x_i^\alpha \, \mathrm{d}i$.

$$\text{Demand for variety } i\text{:} \quad x_i = \left(\frac{\alpha L_Y^{1-\alpha}}{p_i}\right)^{1/(1-\alpha)}, \qquad \varepsilon = \frac{1}{1-\alpha}.$$

$$p_i^* = \frac{1}{\alpha}, \qquad x_i^* = \alpha^{2/(1-\alpha)} L_Y, \qquad \pi = (1-\alpha)\alpha^{\alpha/(1-\alpha)} L_Y.$$

A.5 Romer BGP growth rate

$$g^{\text{mkt}} = \frac{\lambda(1-\alpha)L - \rho}{\sigma + 1}.$$

Scale effect: $\partial g^{\text{mkt}}/\partial L > 0$.

A.6 Solow steady state and Golden Rule

$$\tilde{k}^* = \left[\frac{s}{n + g + \delta}\right]^{1/(1-\alpha)}, \qquad s_{\text{GR}} = \alpha \quad \text{(Cobb-Douglas)}.$$

A.7 Speed of convergence

$$\beta = (1-\alpha)(n + g + \delta).$$

Interpretation: the gap $\hat{k}_t = \ln(\tilde{k}_t/\tilde{k}^*)$ closes at rate β per year.

A.8 Abel et al. (1989) dynamic efficiency test

Economy is dynamically efficient if gross capital income $\geq$ gross investment:

$$\alpha Y \;\geq\; (n + g + \delta)K.$$

A.9 Stone-Geary demand functions

Utility $U = \log(Y_M - \bar{c}_M) + \log Y_S$, income $I = p_M Y_M + p_S Y_S$:

$$Y_M = \bar{c}_M + \frac{I - p_M \bar{c}_M}{2p_M}, \qquad Y_S = \frac{I - p_M \bar{c}_M}{2p_S}.$$

A.10 General R&D technology (ϕ parameter)

$\dot{M} = \lambda M^{\phi} L_A$. On the BGP:

$$\phi = 1: \quad g_M = \lambda L_A; \qquad \phi < 1: \quad g_M = \frac{n}{1-\phi}; \qquad \phi = 0: \quad g_M = \lambda L_A \text{ (requires growing } L_A\text{)}.$$